

JUNE 2020

SOFTWARE TESTING AND QUALITY ASSURANCE

TLEEN BSHARAH

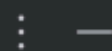
Test Case ID	Test Method Name	Feature Tested	Preconditions	Inputs	Expected Result	Type
TC-01	testValidConstructor	Constructor	Valid initial values	productId="P1"	Object created successfully	Normal
TC-02	testInvalidConstructor	Constructor	Invalid productId	productId=""	IllegalArgumentException	Error
TC-03	testAddStockOverCapacity	addStock	onHand=50, max=100	amount=100	IllegalStateException	Boundary
TC-04	testAddStockNegative	addStock	onHand=50	amount=-5	IllegalArgumentException	Error
TC-05	testReserve	reserve	available=50	amount=10	reserved=10, available=40	Normal
TC-06	testReserveTooMuch	reserve	available=50	amount=100	IllegalStateException	Error
TC-07	testReleaseReservation	releaseReservation	reserved=20	amount=10	reserved=10	Normal
TC-08	testReleaseMoreThanReserved	releaseReservation	reserved=10	amount=20	IllegalStateException	Error
TC-09	testShipReserved	shipReserved	reserved=15	amount=10	onHand=40, reserved=5	Normal

Test Case ID	Test Method Name	Feature Tested	Preconditions	Inputs	Expected Result	Type
TC-10	testShipMoreThanReserved	shipReserved	reserved=5	amount=10	IllegalStateException	Error
TC-11	testRemoveDamaged	removeDamaged	onHand=50	amount=10	onHand=40	Normal
TC-12	testRemoveDamagedTooMuch	removeDamaged	onHand=50	amount=60	IllegalStateException	Error
TC-13	testRemoveDamagedAdjustsReserved	removeDamaged	reserved > onHand	amount=30	reserved adjusted to onHand	Boundary
TC-14	testReorderNeeded	isReorderNeeded	available < threshold	—	returns true	Boundary
TC-15	testReorderNotNeeded	isReorderNeeded	available ≥ threshold	—	returns false	Normal
TC-16	testUpdateReorderThreshold	updateReorderThreshold	Valid threshold	newThreshold=20	threshold updated	Normal
TC-17	testUpdateReorderThresholdNegative	updateReorderThreshold	Invalid input	newThreshold=-1	IllegalArgumentException	Error
TC-18	testUpdateReorderThresholdTooHigh	updateReorderThreshold	Exceeds capacity	newThreshold=200	IllegalArgumentException	Boundary

Test Case ID	Test Method Name	Feature Tested	Preconditions	Inputs	Expected Result	Type
TC-19	testUpdateMaxCapacity	updateMaxCapacity	Valid value	newMax=120	capacity updated	Normal
TC-20	testInvalidMaxCapacity	updateMaxCapacity	newMax < onHand	newMax=10	IllegalStateException	Error
TC-21	testUpdateMaxCapacityEqualOnHand	updateMaxCapacity	newMax = onHand	newMax=50	capacity updated	Boundary
TC-22	testUpdateMaxCapacityInvalid	updateMaxCapacity	Invalid input	newMax=0	IllegalArgumentException	Error
TC-23	testChangeLocation	changeLocation	Valid location	"WH-2-B3"	location updated	Normal
TC-24	testChangeLocationInvalid	changeLocation	Invalid value	""	IllegalArgumentException	Error
TC-25	testGetAvailable	getAvailable	reserved=15	—	available=35	Normal
TC-26	testReserveExactAvailable	reserve	available=50	amount=50	available=0	Boundary
TC-27	testToString	toString	Valid object	—	contains productId	Sanity
TC-28	testReserveInvalid	reserve	amount > available	amount=100	IllegalStateException	Error

Coverage

ProductStockTest ×



Element ^

Class, %

Method, %

Line, %

Branch, %

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all	50% (1/2)	94% (18/19)	80% (62/77)	70% (41/58)
org.example	0% (0/1)	0% (0/1)	0% (0/3)	0% (0/2)
Main	0% (0/1)	0% (0/1)	0% (0/3)	0% (0/2)
ProductStock	100% (1/1)	100% (18/18)	83% (62/74)	73% (41/56)

